Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		Use of $F = Ap$ **and** $A = \pi r^2$ **or** accept $A = 4\pi r^2$ (1) Correct answer = 3 173 [N] (1) [no **ecf** from use of $A = 4\pi r^2$]	1	1		2	2	
	(b)		Fewer collisions …. (1) …because greater distances between molecules (or smaller density or more free space) (1)		2		2		
	(c)	(i)	Application of conservation of energy i.e. $E_k = \frac{Qq}{4\pi\varepsilon_0 r}$ (1) Conversion of 4.7 MeV → J i.e. $4.7 \times 10^6 \times 1.6 \times 10^{-19} = 7.52 \times 10^{-13}$ J (1) Answer = 4.8×10^{-14} [m] (1)		3		3	3	
		(ii)	Smaller than atomic radius or inside plum pudding (1) So force / PE never great enough (for rebound) or scattering angle too large in experiment (1)			2	2		
	(d)		Use of conservation of energy to get speed or momentum e.g. $p^2 = 2mE_k$ etc. $v = 3.75 \times 10^7$ [m s^{-1}] or $p = 3.41 \times 10^{-22}$ [N s] (1) Calculation of a wavelength using $\lambda = \frac{h}{p}$ (even if incorrect, 1.94×10^{-11} m is the correct value) (1) Comparison of the calculated wavelength with atomic separation (or 10^{-9} to 10^{-11} m) (1) Correct final conclusion **and** correct wavelength (1.94×10^{-11} m) (1)			4	4	3	
	(e)		Proton repulsion or like charges repel etc.	1			1		
	(f)		Photon mom calculated $\left(p = \frac{h}{\lambda}\right)$ = 2.73×10^{-22} [kg m s^{-1}] (1) Electron momentum calculated = 9.11×10^{-26} [kg m s^{-1}] (1) [Initial momentum negligible] so final momenta must cancel (1)			3	3	2	